

Suyash Nayan

suyashnyn1@gmail.com | +91 9022498628 | github.com/7suyash7 | linkedin.com/in/suyashnayan

SUMMARY

Software Engineer specialising in **Rust, Go, Java, and C++**. Experienced in Ethereum execution-layer infrastructure, EVM internals, distributed systems, low-latency P2P networking, benchmarking, and backend data pipelines. Proven track record of shipping open-source contributions across protocol clients, MEV infrastructure, and large-scale browser infrastructure.

KEY ACHIEVEMENTS

- **Reduced transaction latency by 61%** in Hyperledger Besu private networks through block-building and transaction-processing optimisation.
- **Shipped 20+ OSS PRs** across **Reth** (Paradigm), **Besu** (Hyperledger), and **Rbuilder** (Flashbots), focused on execution performance, txpool logic, networking, and tooling.
- **Built Rust-based DeFi and network systems** including an N-hop arbitrage solver and a Reth-based P2P mempool crawler with PostgreSQL/WebSocket pipelines.
- **Selected Speaker** at Open Source Summit Japan, Tokyo: [Watch the Talk \(YouTube\)](#) on blockchain performance and benchmarking.

TECHNICAL SKILLS

Languages: Rust, Go, Java, C++, TypeScript/JavaScript, Python, Solidity

Web3/Protocol: EVM Internals, Reth, Hyperledger Besu, Foundry, Hardhat, DeFi AMMs, MEV, RLPx/devp2p, txpool

Backend/Systems: Tokio, Axum, PostgreSQL, WebSockets, Docker, Git, Linux, Benchmarking, Distributed Systems

EXPERIENCE

- **Independent Protocol Engineer & OSS Contributor** Abu Dhabi / Remote
Ethereum Infrastructure, MEV, and DeFi Systems Jan 2024 – Present
 - Built [ArbRs](#), a Rust-based multi-protocol arbitrage engine that simulates Uniswap V3 and Curve swaps locally without external RPC calls.
 - Built a [Reth-based Ethereum P2P mempool crawler](#) to handshake directly with Ethereum mainnet peers, ingest raw transactions, and stream propagation data.
 - Contributed to **Reth** and **Flashbots Rbuilder**, working across execution logic, txpool/networking paths, metrics, CLI tooling, and builder-session lifecycle improvements.
 - Engineered async Rust pipelines using Tokio channels, PostgreSQL indexing, and WebSocket/TUI interfaces for real-time network observability.
- **The Linux Foundation Mentee - Hyperledger** Remote
Protocol Engineer (LFX Fellowship) Jun 2023 – Nov 2023
 - Optimised Hyperledger Besu's block building and transaction processing engine for private consensus networks.
 - Reduced transaction latency by 61% across QBFT, IBFT, and Clique benchmarking environments.
 - Built isolated benchmarking setups to profile throughput, latency, CPU, memory, and disk behaviour under high transaction load.
 - Presented performance findings and optimisation methodology at Open Source Summit Japan in Tokyo.
- **Google Summer of Code - Chromium, Google** Remote
Software Engineer May 2022 – Sep 2022
 - Designed a C++ testing framework for Chromium's rendering engine (Blink), reducing untested drag-and-drop interaction scenarios by 30%.
 - Refactored the Automations API to support consistent cross-browser behaviour across Chrome, Safari, and Firefox.
 - Worked inside a large-scale C++ codebase with strict testing, review, and cross-platform compatibility requirements.
- **CryptoRithm Labs** Remote
Smart Contract Engineer Jan 2022 – Mar 2022
 - Developed an end-to-end test suite for Solidity smart contracts and oracle-backed prediction market flows using Hardhat.
 - Diagnosed and resolved critical smart contract math and state-transition bugs, improving reliability under load testing.
- **Qualyval** Remote
Frontend Engineer Oct 2021 – Dec 2021
 - Modernised legacy React class components into functional components and improved maintainability across the frontend codebase.
 - Implemented serverless monitoring workflows using Puppeteer and scheduled functions for proactive uptime checks.

OPEN SOURCE CONTRIBUTIONS

- **Paradigm: Reth** Rust
• *Open Source Contributor* 2024 – Present
 - Contributed to Ethereum execution-client internals across trie iteration, txpool validation logic, and p2p/networking paths.
 - Worked with async Rust, client architecture, transaction propagation, and performance-sensitive validation code.
- **Hyperledger Besu** Java
• *Core Contributor* 2023
 - Refactored transaction pool and block-building logic to improve propagation speed and block inclusion behaviour.
 - Implemented Spotless fail-fast checks and enhanced JSON-RPC APIs for node administration and developer workflows.
- **Flashbots: Rbuilder** Rust
• *Open Source Contributor* 2024
 - Improved builder tooling, database caching behaviour, system information collection, and builder-session lifecycle management.
 - Added CLI flag support and metrics/tooling improvements for the Rust MEV-Boost block builder.
- **Full OSS Portfolio (All PRs):** [Click here to view Google Doc](#)

PROJECTS

- **ArbRs (Multi-Protocol Arbitrage Engine)** 2025
• *Rust* [Technical Deep Dive](#)
 - Reverse-engineered Uniswap V3 and Curve AMM math into Rust for deterministic local swap simulation.
 - Designed a unified `LiquidityPool` trait to abstract different AMM invariants behind a common pricing interface.
 - Implemented BFS graph search to detect profitable N-hop cycles and model gas/flashloan-adjusted profitability.
- **Reth P2P Mempool Crawler** 2025
• *Rust* [Technical Deep Dive](#)
 - Built a standalone Ethereum node client to connect to mainnet peers and ingest pending transactions directly over devp2p/RLPx.
 - Created a decoupled async pipeline: network ingestion → Tokio MPSC channels → PostgreSQL → WebSocket/TUI visualisation.
 - Indexed live mempool data to analyse transaction propagation, MEV opportunities, and gas price volatility.
- **FlowMark (Load Testing Tool)** 2024
• *Go*
 - Built a Go-based blockchain benchmarking tool to simulate high-concurrency transaction loads and generate performance reports.
 - Secured second place in the Flow Blockchain Hackathon.

EDUCATION

- **Maharashtra Institute of Technology** Pune, India
• *Bachelor of Technology in Computer Science; CGPA: 8.02* Oct 2021 – Oct 2025